

Rising Infant Childcare Costs in the United States: An Exploratory Analysis

Summary of Analysis

This project explores trends in infant childcare costs in the United States using the National Database of Childcare Prices. The analysis focuses on center-based infant childcare costs, represented by the *MCInfant* variable, and examines how median prices changed over time from 2008 to 2018. The dataset includes more than 34,000 records across U.S. states and counties, providing a broad national view of childcare pricing patterns.

Median costs were calculated for each year to understand overall trends rather than focusing on extreme or localized values. Additional visualizations were used to examine geographic patterns by state and region. The results show that infant childcare costs increased substantially over the ten-year period, indicating growing financial pressure on families. This analysis provides a foundation for understanding childcare affordability and highlights areas that may require policy attention.

Findings

Infant childcare is a major expense for families across the United States. From 2008 to 2018, the median cost of center-based infant childcare increased by more than 25 percent. Although there were a few short-term decreases in individual years, the overall trend shows a clear upward trajectory.

Geographic analysis revealed that some states face significantly higher costs than others. Among the top ten states with the highest infant childcare costs, Texas showed the largest aggregated value, followed by Virginia and Illinois. Tennessee had the lowest cost among the top ten states. The geographic dashboard also showed that higher childcare costs are more concentrated along the East Coast, with notable costs in parts of the Midwest and West Coast.

The final years of the dataset, from 2016 to 2018, show a steeper increase compared to earlier years, suggesting that childcare affordability has become a more urgent issue in recent times.

Assumptions

Several assumptions were made during the analysis. It was assumed that the dataset is accurate, complete, and representative of actual childcare costs across states and counties. It was also assumed that the data was collected consistently over time and that no major structural changes in data collection affected the observed trends. Median values were assumed to be an appropriate measure for summarizing childcare costs and minimizing the influence of extreme values.

Items That Still Need Clarification

Further clarification is needed regarding how childcare prices were adjusted, if at all, for inflation. Understanding whether costs are reported in nominal or real terms would help better interpret the magnitude of price increases. Additionally, more detail on how regional cost-of-living differences affect childcare pricing would improve the analysis. Future work could also explore differences between urban and rural areas, as well as variations across counties within the same state.

Direction of Story

The goal of this project is to raise awareness among policymakers about the rising cost of infant childcare and encourage timely policy responses. Increasing childcare costs make it difficult for young families to afford raising children and may contribute to declining birth rates. If left unaddressed, these trends could lead to long-term demographic challenges, including an aging population and labor shortages.

Target Audience

The primary audience for this project includes federal, state, and local policymakers who are responsible for designing family support programs and childcare policies. Secondary audiences include childcare advocacy groups and government agencies involved in workforce and economic planning.

Mediums Included & Why

Three mediums were used in this project: PowerPoint, Power BI, and a policy brief. PowerPoint was used to communicate key findings in a clear and structured format suitable for presentations. Power BI was selected to create interactive dashboards, including maps and charts, that visually highlight geographic differences in childcare costs. The policy brief was included to summarize the issue concisely and present actionable insights for decision-makers.

Design Decisions

The visual design focused on simplicity and clarity, given that the target audience is non-technical. A line chart was used to show changes in median childcare costs over time, as it clearly communicates trends. A clustered bar chart was used to compare costs across the top ten

states, which is difficult to interpret from a line chart alone. A geographic map was included to highlight regional patterns. Neutral colors and clear labeling were chosen to reduce visual distraction and improve readability.

Ethical Considerations

The data used in this project was obtained from a public government source, the National Database of Childcare Prices, which makes it widely accessible and appropriate for academic analysis. No personally identifiable information was included in the dataset, and therefore no individual privacy concerns exist. The data was used in its original form, with no manipulation or removal of observations that could misrepresent results.

Only basic aggregations, such as calculating median values by year or by state, were performed. No data was filtered out without being clearly acknowledged, and no transformations were applied that could distort the findings. There are no known legal or regulatory restrictions associated with the use of this dataset.

One potential risk is that visualizations may oversimplify complex regional differences in childcare costs. To mitigate this risk, multiple visual formats were used, including line charts, bar charts, and geographic maps, to provide context and reduce misinterpretation. Overall, the data was sourced ethically, verified through a credible government database, and presented in a transparent and responsible manner.

Lessons Learned

This project highlighted how rapidly infant childcare costs have increased over time and how significant the financial burden has become for many families. One key lesson learned is the importance of choosing clear and simple visualizations when communicating with non-technical audiences such as policymakers. Effective visuals can make complex data easier to understand and more impactful.

If this project were repeated, additional analysis would be conducted to adjust costs for inflation and to explore differences at the county or metropolitan level. Including comparisons between urban and rural areas would also strengthen the findings. The most enjoyable part of the project was creating the visualizations and seeing how the data clearly illustrated trends that are often discussed but not always quantified.

Rising Infant Childcare Costs in the United States

An Exploratory View Using the National Database of Childcare Prices

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DSC 640 Data Presentation & Visualization
Winter 2025-2026



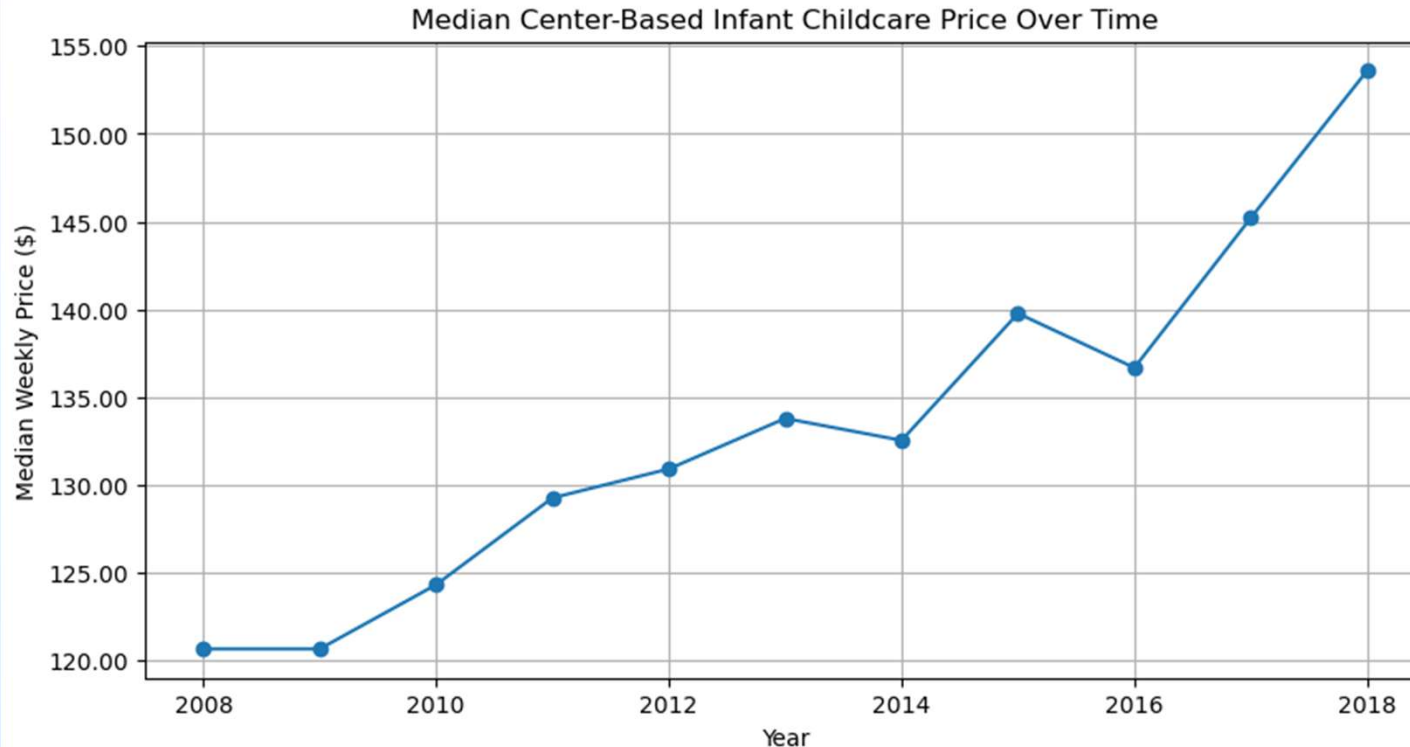
Why This Matters

- Infant childcare is a major expense for families
- Costs have increased over time
- Rising costs can affect family decisions and workforce participation



Childcare Costs Over Time

- Median infant childcare costs increased steadily from 2008 to 2018
- Overall costs rose by more than 25 percent during this period
- While there were small year-to-year drops, the long-term trend is upward



Key Observations

- Overall upward trend across the decade
- Some years show small decreases
- The long-term pattern is increasing costs



Implications for Families

- Increased financial pressure on parents
- Potential impact on birth rates
- Greater burden on lower- and middle-income households



Policy Considerations

- Expand childcare support programs
- Increase childcare supply
- Improve affordability and access



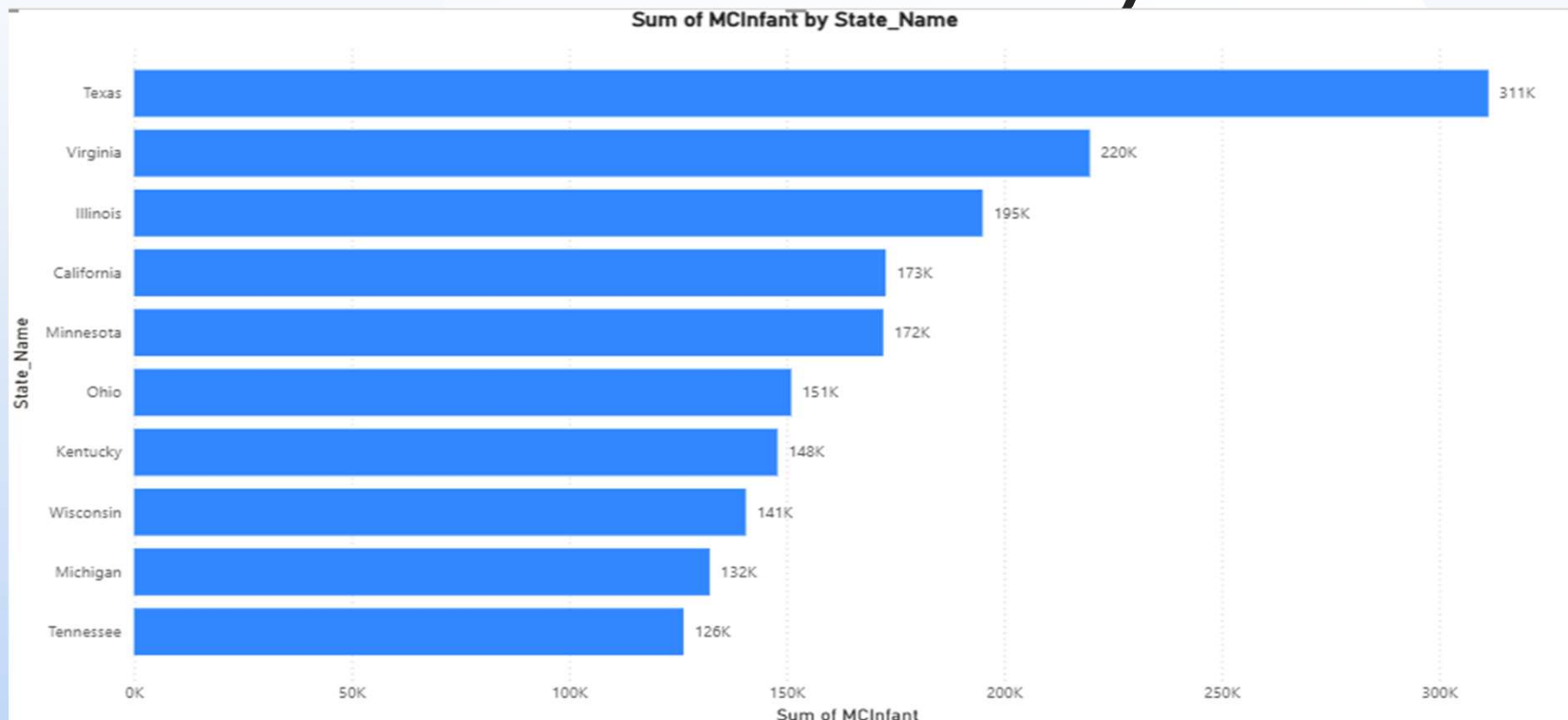
Closing Message



- Childcare affordability is a growing concern
- Data shows sustained increases over time
- Policy attention is needed

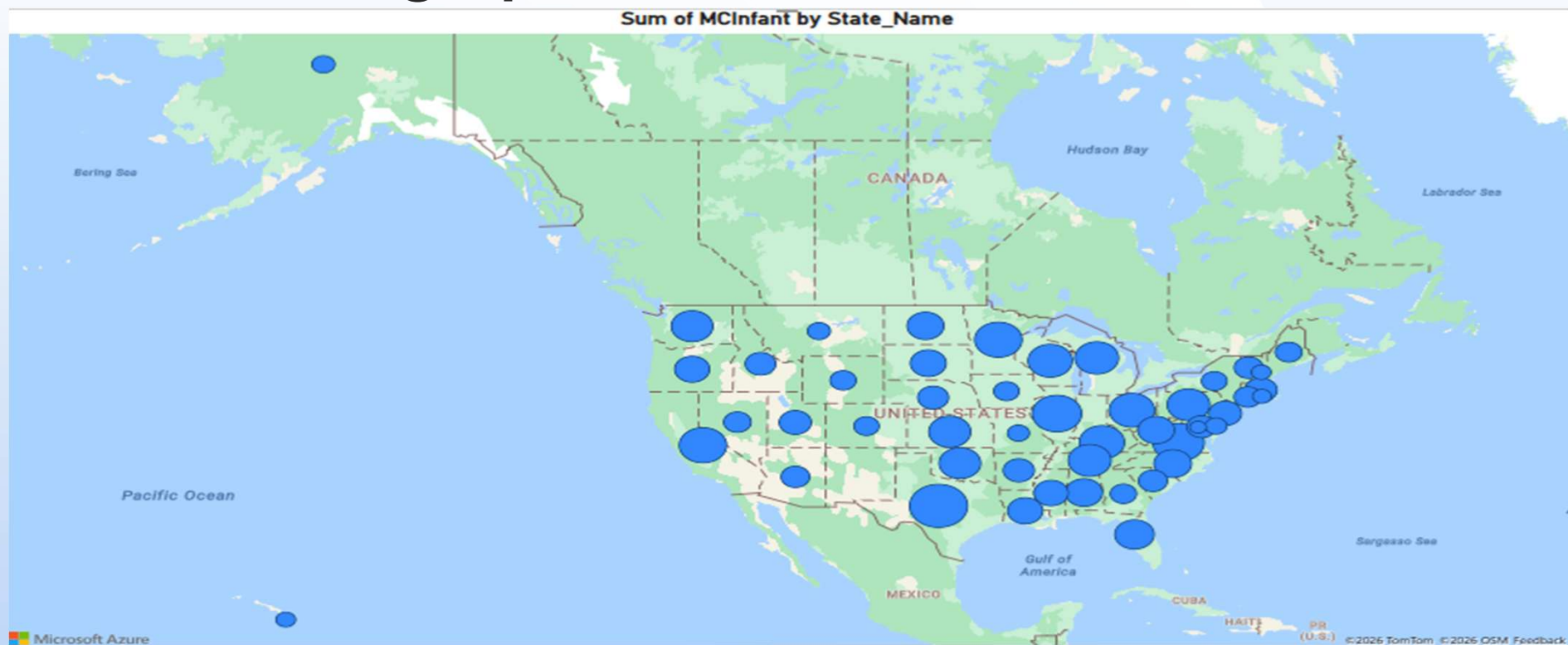


Infant Childcare Costs by State



- The chart highlights the top ten states with the highest infant childcare costs.
- There is a significant gap between the highest and lowest states within the top ten group.
- States such as Texas, Virginia, and Illinois show particularly high-cost levels compared to others.

Geographic Distribution of Costs



- The map shows how infant childcare costs are distributed across the United States.
- Larger circles represent higher overall cost levels.
- Higher cost concentrations appear in several coastal states compared to many central states.
- The map complements the bar chart by showing where high-cost states are located, not just how they rank.

Design Narrative

For this project, specific charts and dashboard visuals were chosen to clearly communicate the story of rising infant childcare costs. A line chart was selected to display childcare costs over time from 2008 to 2018. This chart shows a single line changing over time, which makes it easy for non-technical audiences to understand how prices have increased. Line charts are also commonly used in presentations, so for an audience such as policymakers and childcare lawmakers, this format is familiar and easy to interpret.

A geographic dashboard was used to display a map of the United States with circles that vary in size by childcare cost. Larger circles represent higher childcare costs, making it easy to see where families face greater financial pressure. This visualization allows policymakers and government agencies to quickly identify states where rising childcare costs may require more urgent policy attention or support programs.

A clustered bar chart was also included in the dashboard to show the top ten states with the highest infant childcare costs. While the geographic map provides a broad regional view, it can be difficult to determine exact rankings. The bar chart clearly displays states in order from highest to lowest cost, making comparisons more direct. For example, it becomes clear which states have the highest costs and which fall at the lower end of the top ten.

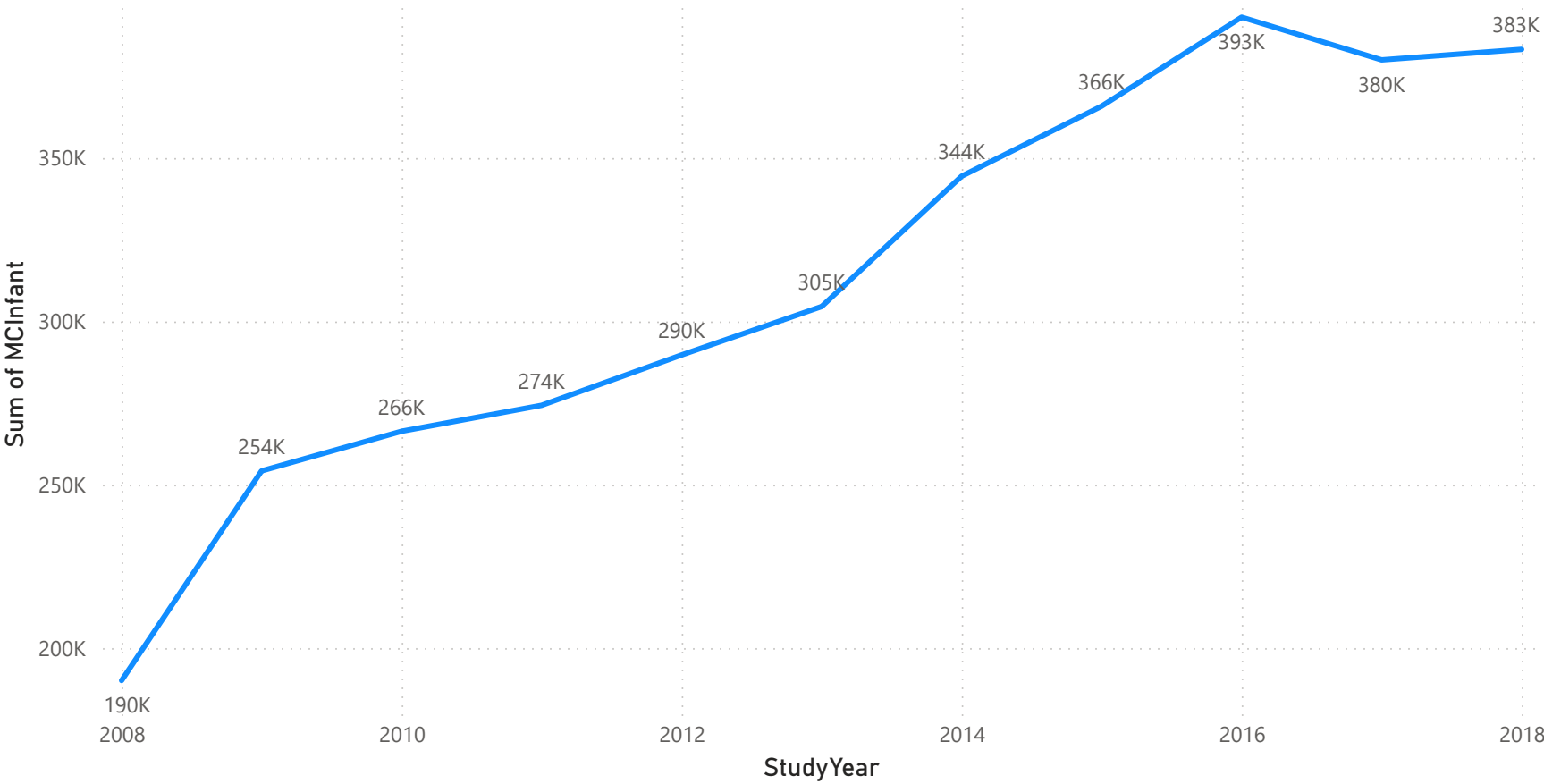
Simple color choices were used throughout the visuals to avoid distraction. Blue bars and lines on a white background were selected for readability and clarity. For the map, a green background with blue circles was used to keep the design clean and easy to follow. Overall, this design fits the policymaker audience because it allows viewers to quickly understand both long-term trends and regional differences without needing technical expertise. The combination of charts works together, with each visual highlighting information that the others cannot show on their own.

Infant Childcare Cost Dashboard

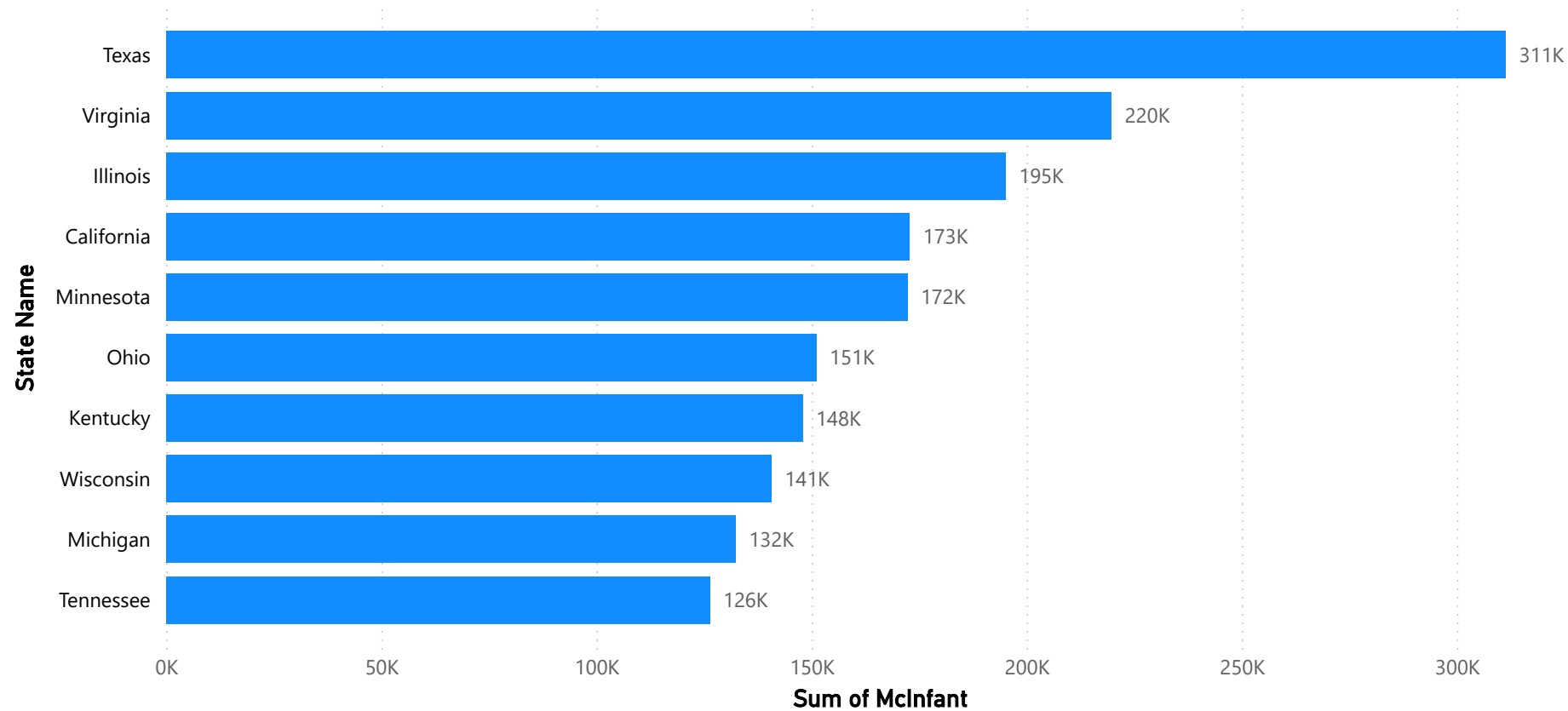
National Database of Childcare Prices (2008–2018)

Created using Microsoft Power BI

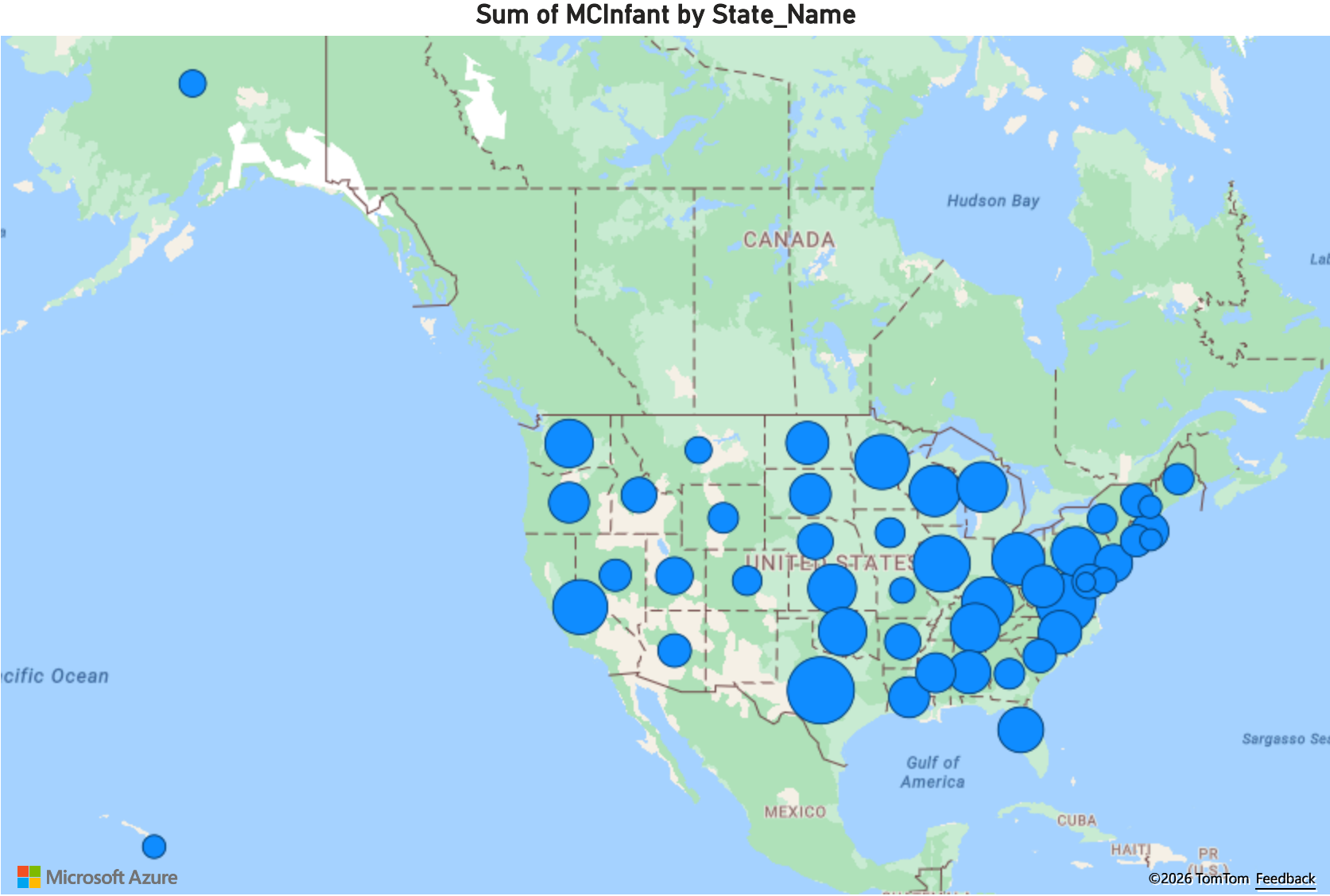
Trend in Median Infant Childcare Costs (2008–2018)



Top Ten States by Infant Childcare Cost



Geographic Distribution of Infant Childcare Costs



Policy Brief

This one-page policy brief is designed for policymakers and legislative agencies to clearly communicate the growing issue of rising infant childcare costs and to encourage targeted policy action. Between 2008 and 2018, median infant childcare costs increased by more than 25 percent, placing significant financial pressure on young families and individuals considering starting a family. As childcare expenses continue to rise, many parents are forced to limit family size or delay having children due to affordability concerns.

The policy brief will highlight how sustained increases in childcare costs may contribute to broader economic and workforce challenges, particularly as the population ages and fewer young workers enter the labor market. The brief will emphasize the importance of proactive policy intervention to support families and maintain long-term economic stability. Proposed policy considerations will include expanding childcare assistance programs, investing in additional infant childcare centers in high-cost regions, and implementing targeted tax benefits or subsidies to reduce the financial burden on lower- and middle-income households.